

Location-invariant Multi-sample U -tests for Covariance Matrices with Large Dimension

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ABSTRACT. For two or more multivariate distributions with common covariance matrix, test statistics for certain special structures of the common covariance matrix are presented when the dimension of the multivariate vectors may exceed the number of such vectors. The test statistics are constructed as functions of location-invariant estimators defined as U -statistics, and the corresponding asymptotic theory is used to derive the limiting distributions of the proposed tests. The properties of the test statistics are established under mild and practical assumptions, and the same are numerically demonstrated using simulation results with small or moderate sample sizes and large dimensions.

Key words: high-dimensional inference, multi-sample sphericity, U -statistics

1. Introduction

Consider a data setup from $g \geq 2$ independent multivariate populations, that is,

$$\mathbf{X}_{ik} = (X_{i1k}, \dots, X_{ipk})' \sim \mathcal{F}_i,$$

$k = 1, \dots, n_i$, are independent vectors drawn from the i th population with distribution function $\mathcal{F}_i$ with $E(\mathbf{X}_{ik}) = \boldsymbol{\mu}_i \in \mathbb{R}^p$, $\text{Cov}(\mathbf{X}_{ik}) = \boldsymbol{\Sigma}_i > 0$, $i = 1, \dots, g$. Assuming homogeneity of covariance matrices across g populations, that is, $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma} \forall i$, we are interested to test certain hypotheses concerning the common $\boldsymbol{\Sigma}$ when the dimension p may exceed the sample sizes, that is, $p \gg n_i$, and $\mathcal{F}_i$ need not necessarily be multivariate normal. In particular, we are interested to test

$$H_{0a} : \boldsymbol{\Sigma} \propto \mathbf{I} \quad \text{and} \quad H_{0b} : \boldsymbol{\Sigma} = \mathbf{I} \quad (1)$$

as multi-sample sphericity and identity hypotheses of $\boldsymbol{\Sigma}$. Note that, H_{0b} pertains to testing any arbitrary structure of $\boldsymbol{\Sigma}$, say $\boldsymbol{\Sigma}_0 > 0$.

Because the number of parameters to be tested drastically reduces under the two hypotheses in (1), they are some of the most frequently tested ones in multivariate theory. Of particular mention is the case of multi-factor or multi-sample repeated measures designs where multi-sample sphericity is a part of the assumptions underlying the model (Crowder & Hand, 1990; Davis, 2002; Timm, 2002). In fact, the applications in repeated measures designs give it a special place in psychological studies; see, for example, Harris (1984), Winer *et al.* (1991), Mendoza *et al.* (1976) and Mendoza (1980). Another important application is in non-hierarchical clustering using mixture of distributions (Johnson & Wichern, 2007). Further, the tests of (1) are of special value as preliminary tests to simplify the mainstream analysis, for example, in multivariate analysis of variance or discriminant analysis.

For the classical $p < n_i$ case, there exist many useful tests for (1); see, for example, Anderson (2003), Seber (2004) and Muirhead (2005). For the high-dimensional case, the likelihood-ratio tests collapse and need to be modified. Several attempts have recently been made in this direction, although most of them are one-sample modifications and based on the classical tests for

H_{0a} and H_{0b} originally given, respectively, in John (1971) and Nagao (1973); see, for example, Ledoit & Wolf (2002), Chen *et al.* (2010) and Ahmad (2014b); see also Fujikoshi *et al.* (2010).

Our aim is to introduce tests for (1) given that the homogeneity hypothesis across g populations already holds. For homogeneity tests under a similar setup as used here, see, for example, Li & Chen (2012) and Ahmad (2014b). By pooling information from all g samples, we first define location-invariant estimators as U -statistics so that they are not only unbiased but also possess optimality property (Polansky, 2011, p. 482), and their consistency is shown to be intact under high-dimensional asymptotics. Using these estimators, we define tests of H_{0a} and H_{0b} , which can be used for very small sample sizes and arbitrary large dimension. Moreover, the distribution-free estimators also help us to easily extend the tests to a more general class of multivariate models.

The proposed tests are also based on John and Nagao's one-sample tests, but the approach differs in many respects from the others. Unlike the aforementioned one-sample tests, we present an extension to the multi-sample case. Further, the tests are defined for a flexible general multivariate model, including normality, that allows us to study the properties of the tests under only a few, mild assumptions. Asymptotically, both tests reduce to the same form so that their limit distributions can be derived simultaneously. Moreover, these limiting distributions are derived using the asymptotic theory of U -statistics and the properties of their kernels, which are degenerate under the null and non-degenerate under the alternative.

After a detailed treatment of two-sample tests under normality in Section 2, the multi-sample case is discussed in Section 3. Section 4 extends both cases to a general multivariate model. A simulation study of the tests is given in Section 5. Section 6 summarizes the main results. All proofs are in the appendix.

2. Test statistics for two populations

For $g = 2$, let $\mathbf{X}_{1k}$ and $\mathbf{X}_{2l}$ be $p \times 1$ independent random vectors where

$$\mathbf{X}_{1k} \sim \mathcal{N}_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1), \quad \mathbf{X}_{2l} \sim \mathcal{N}_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2), \quad (2)$$

$\boldsymbol{\Sigma}_i > 0, i = 1, 2$. Under homogeneity, $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma}$, we are interested to test H_{0a} and H_{0b} . Our construction of the test statistics is based on a modification of the corresponding one-sample test statistics in the classical case as given in John (1971) and Nagao (1973) for H_{0a} and H_{0b} , using the equations

$$\frac{1}{p} \text{tr} \left(\frac{\boldsymbol{\Sigma}}{\frac{1}{p} \text{tr}(\boldsymbol{\Sigma})} - \mathbf{I} \right)^2 = \frac{\frac{1}{p} \text{tr}(\boldsymbol{\Sigma}^2)}{\left[\frac{1}{p} \text{tr}(\boldsymbol{\Sigma}) \right]^2} - 1 = \frac{p \text{tr}(\boldsymbol{\Sigma}^2)}{[\text{tr}(\boldsymbol{\Sigma})]^2} - 1 \quad (3)$$

$$\frac{1}{p} \text{tr}(\boldsymbol{\Sigma} - \mathbf{I})^2 = \frac{1}{p} \text{tr}(\boldsymbol{\Sigma}^2) - \frac{2}{p} \text{tr}(\boldsymbol{\Sigma}) + 1. \quad (4)$$

The modification is based on defining moment estimators of the three traces, $\text{tr}(\boldsymbol{\Sigma})$, $\text{tr}(\boldsymbol{\Sigma}^2)$ and $[\text{tr}(\boldsymbol{\Sigma})]^2$, and using them directly as plug-in estimators in the aforementioned equations. These trace estimators will be so defined that the resulting test statistics are approximately normally distributed with their accuracy intact even when p far exceeds n_i . For the present case, we define the pooled plug-in estimators of the three traces by combining information from both independent samples. Let C_1 , C_2 and C_3 denote these pooled estimators of $\text{tr}(\boldsymbol{\Sigma})$, $\text{tr}(\boldsymbol{\Sigma}^2)$

and $[\text{tr}(\Sigma)]^2$, respectively. Then, using them as plug-in estimators in equations (3) and (4), the test statistics for H_{0a} and H_{0b} can be defined, respectively, as

$$T_a = \frac{pC_3}{C_2} - 1 \quad \text{and} \quad T_b = \frac{1}{p}C_3 - \frac{2}{p}C_1 + 1. \quad (5)$$

To define C_1, C_2 and C_3 , we first need to define estimators of the traces from individual samples. There are obviously many ways to construct estimators of these traces with several desired properties such as unbiasedness and consistency. We aim to define these estimators as U -statistics so that they are not only unbiased but also consistent for both large n and large p . Note that the main component in all traces is the unknown common covariance matrix Σ . Consider now the first sample with vector $\mathbf{X}_{1k}$, with its classical unbiased estimator $\hat{\Sigma}_1$ defined as $\hat{\Sigma}_1 = \sum_{k=1}^{n_1} (\mathbf{X}_{1k} - \bar{\mathbf{X}}_1)(\mathbf{X}_{1k} - \bar{\mathbf{X}}_1)' / (n_1 - 1)$ with $\bar{\mathbf{X}}_1 = \sum_{k=1}^{n_1} \mathbf{X}_{1k} / n_1$ as the sample mean vector; similarly, we can define $\hat{\Sigma}_2$.

From the theory of U -statistics, however, $\hat{\Sigma}_1$ and $\hat{\Sigma}_2$ can also be defined as second-order U -statistics. Let $\mathbf{D}_{1kr} = \mathbf{X}_{1k} - \mathbf{X}_{1r}, k \neq r$ and $\mathbf{D}_{2ls} = \mathbf{X}_{2l} - \mathbf{X}_{2s}, l \neq s$ be within-sample pairwise differences from samples 1 and 2, respectively, with $E(\mathbf{D}_{1kr}) = 0 = E(\mathbf{D}_{2ls})$ and $\text{Cov}(\mathbf{D}_{1kr}) = E(\mathbf{D}_{1kr}\mathbf{D}_{1kr}') = 2\Sigma_1$, $\text{Cov}(\mathbf{D}_{2ls}) = E(\mathbf{D}_{2ls}\mathbf{D}_{2ls}') = 2\Sigma_2$. Then, the U -statistics

$$\hat{\Sigma}_1 = \frac{1}{Q(n_1)} \sum_{k=1}^{n_1} \sum_{r=1, r \neq k}^{n_1} \frac{1}{2} \mathbf{D}_{1kr} \mathbf{D}_{1kr}' \quad \text{and} \quad \hat{\Sigma}_2 = \frac{1}{Q(n_2)} \sum_{l=1}^{n_2} \sum_{s=1, s \neq l}^{n_2} \frac{1}{2} \mathbf{D}_{2ls} \mathbf{D}_{2ls}',$$

$Q(n_i) = n_i(n_i - 1)$, are unbiased estimators of $\Sigma_i, i = 1, 2$. With $E(\hat{\Sigma}_i) = \Sigma_i$ and thus $E[\text{tr}(\hat{\Sigma}_i)] = \text{tr}(\Sigma_i)$, the corresponding unbiased estimators of $\text{tr}(\Sigma_1)$ and $\text{tr}(\Sigma_2)$ can be defined, again as U -statistics, as

$$B_1 = \frac{1}{Q(n_1)} \sum_{k=1}^{n_1} \sum_{r=1, r \neq k}^{n_1} \frac{1}{2} A_{1kr} \quad \text{and} \quad B_2 = \frac{1}{Q(n_2)} \sum_{l=1}^{n_2} \sum_{s=1, s \neq l}^{n_2} \frac{1}{2} A_{2ls}, \quad (6)$$

where $A_{1kr} = \mathbf{D}_{1kr}' \mathbf{D}_{1kr}$ and $A_{2ls} = \mathbf{D}_{2ls}' \mathbf{D}_{2ls}$. Note that, the symmetrical kernels $h(\mathbf{X}_{1k}, \mathbf{X}_{1r}) = A_{1kr}/2$ and $h(\mathbf{X}_{2l}, \mathbf{X}_{2s}) = A_{2ls}/2$, ensure unbiasedness of the estimators. Using the independence of the two samples, we also note that $E(\hat{\Sigma}_1 \hat{\Sigma}_2) = \Sigma_1 \Sigma_2$ so that $E[\text{tr}(\hat{\Sigma}_1 \hat{\Sigma}_2)] = \text{tr}(\Sigma_1 \Sigma_2)$. Then, with $\text{tr}(\mathbf{D}_{1kr} \mathbf{D}_{1kr}' \mathbf{D}_{2ls} \mathbf{D}_{2ls}') = \mathbf{D}_{2ls}' \mathbf{D}_{1kr} \mathbf{D}_{1kr}' \mathbf{D}_{2ls} = (\mathbf{D}_{1kr}' \mathbf{D}_{2ls})^2 = A_{12krls}^2$, an unbiased estimator of $\text{tr}(\Sigma_1 \Sigma_2)$ can be defined as

$$B_{12} = \frac{1}{P(n_1, n_2)} \sum_{k=1}^{n_1} \sum_{r=1, r \neq k}^{n_1} \sum_{l=1}^{n_2} \sum_{s=1, s \neq l}^{n_2} \frac{1}{4} A_{12krls}^2, \quad (7)$$

where $P(n_1, n_2) = Q(n_1)Q(n_2)$. Note that, B_{12} is a two-sample U -statistic with kernel $h(\mathbf{X}_{1k}, \mathbf{X}_{1r}, \mathbf{X}_{2l}, \mathbf{X}_{2s}) = A_{12krls}^2/4$. It gives an unbiased estimator of $\text{tr}(\Sigma^2)$ when $\Sigma_1 = \Sigma_2 = \Sigma$. Obviously, like B_i for $\text{tr}(\Sigma_i)$, similar within-sample independent estimators can also be defined to construct a pooled estimator of $\text{tr}(\Sigma^2)$. Such estimators are discussed in Ahmad (2014b) in the context of testing homogeneity of covariance matrices, where it is argued that they are heavy in terms of n_i and give slower convergence as compared with estimators like B_{12} .

Further, B_i and B_{12} are location-invariant, leading to the proposed tests also location-invariant, so that we can assume, without loss of generality, $\mu_i = \mathbf{0}$. However, similar U -statistics-based non-invariant estimators can also be defined for the same traces. They are

simpler, but they depend on the assumption of zero mean vectors. Moreover, B_i and B_{12} being U -statistics help us employ the corresponding asymptotic theory to arrive at the limit distributions of the test statistics. Additionally, the optimality (minimum variance) property of U -statistics (Polansky, 2011, Theorem 11.3) help us ensure same properties for the estimators and hence the test statistics.

To effectively use these estimators further, particularly to arrive at the limiting distributions of the test statistics, we need the following assumptions.

Assumption 1. For $p \rightarrow \infty$, let $\frac{\text{tr}(\Sigma)}{p} = O(1)$.

Assumption 2. $\inf_p \frac{\text{tr}(\Sigma^4)}{[\text{tr}(\Sigma^2)]^2} > 0$.

Assumption 3. For $n_i, p \rightarrow \infty$, $\frac{p}{n_i} \rightarrow c \in (0, \infty)$, $i = 1, \dots, g$.

An immediate consequence of assumption 1 is that $\text{tr}(\Sigma^2)/p^2 = O(1)$, so that a reference to assumption 1 in the sequel may also imply its consequence. Assumption 1 is mild and frequently used in high-dimensional inference. Assumption 2 deviates from its usual form in the literature where such trace ratios are assumed to vanish. It can be shown that assumption 2 holds for commonly used covariance structures as $p \rightarrow \infty$ (Ahmad, 2014a). All assumptions are only required for the distributions under the alternative. Under the null, assumption 1 holds trivially, whereas assumption 2 neither holds nor is required; similarly, assumption 3 is also not needed.

If we let $\lambda_s, s = 1, \dots, p$, be the eigenvalues of common Σ and correspondingly $\nu_s = \lambda_s/p$ be the eigenvalues of the scaled Σ/p , then the assumptions refer to moments of ν_s . In the proof of our main result (theorem 3), it is shown that these eigenvalues eventually emerge in the limits as the eigenvalues of the kernels of U -statistics. In fact, under the null, the kernels are degenerate and, by square-integrability under the consequence of assumption 1, Hilbert–Schmidt kernels. By the spectral decomposition of this kernel, the non-degenerate limit under the null only comes indirectly, unlike under the alternative where the non-degenerate normal limit follows directly. To follow this plan, we begin with the moments of estimators in the following theorem, proved in Appendix B1.

Theorem 1. Let B_i and B_{12} be the estimators of $\text{tr}(\Sigma_i)$ and $\text{tr}(\Sigma_1 \Sigma_2)$, $i = 1, 2$, defined in equations (6) and (7). Let $n = n_1 + n_2$. Then, under model (2),

$$E(B_i) = \text{tr}(\Sigma_i), \quad E(B_{12}) = \text{tr}(\Sigma_1 \Sigma_2) \quad (8)$$

$$\text{Var}(B_i) = \frac{2}{n_i - 1} \text{tr}(\Sigma_i^2) \quad (9)$$

$$\text{Var}(B_{12}) = \frac{2}{(n_1 - 1)(n_2 - 1)} \left[(n - 1) \text{tr}\{(\Sigma_1 \Sigma_2)^2\} + [\text{tr}(\Sigma_1 \Sigma_2)]^2 \right] \quad (10)$$

$$\text{Cov}(B_i, B_{12}) = \frac{2}{n_i - 1} \text{tr}(\Sigma_i^2 \Sigma_j), \quad i = 1, 2, i \neq j; \quad \text{Cov}(B_1, B_2) = 0. \quad (11)$$

Further, it follows that $\text{Var}(B_i/E(B_i)) = O(1/n_i) = \text{Cov}(B_i/E(B_i), B_{12}/E(B_{12}))$ and $\text{Var}(B_{12}/E(B_{12})) = O(1/n_1 + 1/n_2)$, $i = 1, 2$.

Following theorem 1, the estimators are unbiased and consistent, and their moment-ratios are uniformly bounded in p . This last property will help us establish similar bounds for the pooled estimators of the traces, which will further be exploited to determine the asymptotic distributions of the test statistics. For now, the following theorem adds another property to B_i and B_{12} , namely, asymptotic normality. Let $\tilde{B}_i = B_i/E(B_i) - 1$, $i = 1, 2$, $\tilde{B}_{12} = B_{12}/E(B_{12}) - 1$. Then, $\tilde{B}_i$ and $\tilde{B}_{12}$ have zero means. For proof, see Appendix B2.

Theorem 2. Given theorem 1, let $\tilde{B}_i$, $i = 1, 2$ and $\tilde{B}_{12}$ be as defined earlier. Then, under model (2) and assumption 1, as $n_i, p \rightarrow \infty$, $i = 1, 2$,

$$\tilde{B}_i/\sqrt{\text{Var}(\tilde{B}_i)} \xrightarrow{D} N(0, 1), \quad \tilde{B}_{12}/\sqrt{\text{Var}(\tilde{B}_{12})} \xrightarrow{D} N(0, 1). \quad (12)$$

Now, we define C_1 , C_2 and C_3 , as pooled estimators of $\text{tr}(\Sigma)$, $[\text{tr}(\Sigma)]^2$ and $\text{tr}(\Sigma^2)$, as

$$C_1 = \frac{Q(n_1)B_1 + Q(n_2)B_2}{Q(n_1) + Q(n_2)} \\ = \frac{1}{Q(n_1) + Q(n_2)} \left(\sum_{k=1}^{n_1} \sum_{r=1}^{n_1} \frac{1}{2} A_{1kr} + \sum_{l=1}^{n_2} \sum_{s=1}^{n_2} \frac{1}{2} A_{2ls} \right) \quad (13)$$

$$C_2 = B_1 B_2 = \frac{1}{Q(n_1)Q(n_2)} \sum_{k=1}^{n_1} \sum_{r=1}^{n_1} \sum_{l=1}^{n_2} \sum_{s=1}^{n_2} \frac{1}{4} A_{1kr} A_{2ls} \quad (14)$$

$$C_3 = B_{12}, \quad (15)$$

with B_{12} as in equation (7). In Section 3, we extend T_a and T_b for the general case by extending C_1 , C_2 and C_3 to g samples and state their properties in theorem 4. The results of this theorem, for $g = 2$ with $n = n_1 + n_2$, reduce as follows.

$$\text{Var}(C_1) = \frac{n_1 Q(n_1) + n_2 Q(n_2)}{[Q(n_1) + Q(n_2)]^2} 2\text{tr}(\Sigma^2) \quad (16)$$

$$\text{Var}(C_2) = \frac{2\text{tr}(\Sigma^2)}{(n_1 - 1)(n_2 - 1)} \left[2\text{tr}(\Sigma^2) + (n - 2)[\text{tr}(\Sigma)]^2 \right] \quad (17)$$

$$\text{Var}(C_3) = \text{Var}(B_{12}) \quad (18)$$

$$\text{Cov}(C_1, C_2) = \frac{2n}{Q(n_1) + Q(n_2)} \text{tr}(\Sigma^2) \text{tr}(\Sigma) \quad (19)$$

$$\text{Cov}(C_1, C_3) = \frac{2n}{Q(n_1) + Q(n_2)} \text{tr}(\Sigma^3) \quad (20)$$

$$\text{Cov}(C_2, C_3) = \frac{4}{P(n_1, n_2)} \left[\text{tr}(\Sigma^4) + [Q(n_1) + Q(n_2) + 1] \text{tr}(\Sigma^3) \text{tr}(\Sigma) \right]. \quad (21)$$

Clearly, $\text{Var}(C_i/E(C_i)) = O(1/n_1 + 1/n_2)$, $i = 1, 2$, $\text{Var}(C_3/E(C_3)) = O(1/n_1 n_2)$, $\text{Cov}(C_1/\text{tr}(\Sigma), C_2/[\text{tr}(\Sigma)]^2) = O(1/n_1 + 1/n_2)$ and the other two covariance ratios are

$O(1/n_1^2 + 1/n_2^2)$. To see how these bounds help us to conveniently establish the properties and asymptotic distributions of the proposed tests, using only a few mild assumptions, we rewrite the test statistics from (5) as

$$\frac{T_a + 1}{\psi} = \frac{pC_3}{C_2} \cdot \frac{[\text{tr}(\mathbf{\Sigma})]^2}{p\text{tr}(\mathbf{\Sigma}^2)} = \frac{C_3/\text{tr}(\mathbf{\Sigma}^2)}{C_2/[\text{tr}(\mathbf{\Sigma})]^2} \quad (22)$$

$$\begin{aligned} T_b - \frac{1}{p}\text{tr}(\mathbf{\Sigma} - \mathbf{I})^2 &= \frac{1}{p}[C_3 - \text{tr}(\mathbf{\Sigma}^2)] - \frac{2}{p}[C_1 - \text{tr}(\mathbf{\Sigma})] \\ &= \frac{\text{tr}(\mathbf{\Sigma}^2)}{p} \left(\frac{C_3}{\text{tr}(\mathbf{\Sigma}^2)} - 1 \right) - \frac{2\text{tr}(\mathbf{\Sigma})}{p} \left(\frac{C_1}{\text{tr}(\mathbf{\Sigma})} - 1 \right), \end{aligned} \quad (23)$$

with $\psi = \frac{p\text{tr}(\mathbf{\Sigma}^2)}{[\text{tr}(\mathbf{\Sigma})]^2}$ estimated as $\hat{\psi} = \frac{pC_3}{C_2}$. Note that the tests are now scaled by the traces they estimate. The aforementioned bounds of moment-ratios are for these scaled tests. From the moments and bounds of C_1 and C_2 , it immediately follows that $C_1/\text{tr}(\mathbf{\Sigma}) \xrightarrow{P} 1$, $C_2/[\text{tr}(\mathbf{\Sigma})]^2 \xrightarrow{P} 1$, as $n_1, n_2, p \rightarrow \infty$. Then, T_a and T_b are functions of C_3 alone, and their variances come from $\text{Var}(C_3)$, that is,

$$T_a = \frac{C_3}{\text{tr}(\mathbf{\Sigma}^2)} - 1 = T_b \Rightarrow \text{Var}(T_a) = \frac{\text{Var}(C_3)}{[\text{tr}(\mathbf{\Sigma}^2)]^2} = \text{Var}(T_b), \quad (24)$$

ignoring the vanishing $o_P(1)$ terms. This not only simplifies the tests but also helps determine their limiting distributions and, for both tests simultaneously, as given in the following theorem, proved in Appendix B3.

Theorem 3. *Let T_a and T_b be as defined in equations (22) and (23), respectively. Then, under model (2) and assumptions 1–3, as $n_i, p \rightarrow \infty$,*

$$\begin{aligned} \sigma_{T_a}^{-1} \left(\frac{T_a + 1}{\psi} - 1 \right) &\xrightarrow{D} N(0, 1) \\ \sigma_{T_b}^{-1} \left[T_b - \frac{1}{p}\text{tr}(\mathbf{\Sigma} - \mathbf{I})^2 \right] &\xrightarrow{D} N(0, 1), \end{aligned}$$

where $\sigma_{T_a}^2 = \sigma_{T_b}^2$ denote the variances of T_a and T_b . In particular, under H_{0j} and Assumption 1, $\sqrt{n_1 n_2 / 2} T_j \xrightarrow{D} N(0, 1)$, $j = a$ or $j = b$.

3. Multi-sample case

With detailed explanations already given for $g = 2$, an extension to $g \geq 2$ is simple. We, therefore, only present main results, avoiding unnecessary details and proofs. Let

$$\mathbf{X}_{ik} = (X_{ik1}, \dots, X_{ikp})' \sim \mathcal{N}_p(\boldsymbol{\mu}_i, \mathbf{\Sigma}_i), \quad i = 1, \dots, g. \quad (25)$$

First, the estimators B_i, B_{ij} , in their general form, can be expressed as

$$B_i = \frac{1}{Q(n_i)} \sum_{k=1}^{n_i} \sum_{\substack{r=1 \\ k \neq r}}^{n_i} \frac{1}{2} A_{ikr} \quad (26)$$

$$B_{ij} = \frac{1}{P(n_i, n_j)} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{r=1}^{n_i} \sum_{\substack{l=1 \\ l \neq s}}^{n_j} \sum_{s=1}^{n_j} \frac{1}{4} A_{ijklrs}^2, \quad (27)$$

$A_{ikr} = \mathbf{D}'_{ikr} \mathbf{D}_{ikr}$, $A_{ijkrls} = \mathbf{D}'_{ikr} \mathbf{D}_{jls}$, $Q(n_i) = n_i(n_i - 1)$, $P(n_i, n_j) = Q(n_i)Q(n_j)$. Their properties in theorems 1 and 2 remain the same by replacing the subscript 1 with i and subscript 2 with j . T_a and T_b follow again from equations (22) and (23), assuming that $\Sigma_i = \Sigma \forall i$. Now, we pool information from g samples and define multi-sample versions of equations (13)–(15), where $P^* = \sum_{i \neq j}^g P(n_i, n_j)$.

$$C_1 = \frac{1}{\sum_{i=1}^g Q(n_i)} \sum_{i=1}^g Q(n_i) B_i \quad (28)$$

$$C_2 = \frac{1}{g(g-1)} \sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g B_i B_j \quad (29)$$

$$C_3 = \frac{1}{P^*} \sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g P(n_i, n_j) B_{ij}, \quad (30)$$

The moments of C_1 , C_2 and C_3 and their bounds are given in theorem 4 and corollary 1, respectively. For proofs, see Appendix B4.

Theorem 4. Let C_1 , C_2 and C_3 be as defined in equations (28)–(30), respectively. Let $n_{ij} = n_i + n_j$ and $g^* = g(g-1)(g-2)$. Then, under model (25),

$$E(C_1) = \text{tr}(\Sigma), \quad E(C_2) = [\text{tr}(\Sigma)]^2, \quad E(C_3) = \text{tr}(\Sigma^2) \quad (31)$$

$$\text{Var}(C_1) = \frac{2\text{tr}(\Sigma^2)}{[\sum_{i=1}^g Q(n_i)]^2} \sum_{i=1}^g n_i Q(n_i) \quad (32)$$

$$\begin{aligned} \text{Var}(C_2) = & \frac{8\text{tr}(\Sigma^2)}{g^2(g-1)^2} \left(\sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g \frac{4}{(n_i-1)(n_j-1)} \text{tr}(\Sigma^2) \right. \\ & \left. + (g-1)^2 [\text{tr}(\Sigma)]^2 \sum_{i=1}^g \frac{1}{n_i-1} \right) \end{aligned} \quad (33)$$

$$\begin{aligned} \text{Var}(C_3) = & \frac{4}{P^{*2}} \left(\sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g P(n_i, n_j) \left[n_i n_j \left\{ (n_{ij}-1) \text{tr}(\Sigma^4) + [\text{tr}(\Sigma^2)]^2 \right\} \right] \right. \\ & \left. + g^* \left[32\text{tr}(\Sigma^4) + 15[\text{tr}(\Sigma^2)]^2 \right] \right) \end{aligned} \quad (34)$$

$$\text{Cov}(C_1, C_2) = \frac{4 \sum_{i=1}^g n_i}{g \sum_{i=1}^g Q(n_i)} \text{tr}(\Sigma^2) \text{tr}(\Sigma) \quad (35)$$

$$\text{Cov}(C_1, C_3) = \frac{4 \sum_{i=1}^g n_i}{\sum_{i=1}^g Q(n_i)} \text{tr}(\Sigma^3) \quad (36)$$

$$\text{Cov}(C_2, C_3) = \frac{8}{P^*} \left(\text{tr}(\Sigma^4) + \text{tr}(\Sigma^3) \text{tr}(\Sigma) + \text{tr}(\Sigma^3) \text{tr}(\Sigma) \sum_{i=1}^g Q(n_i) \right). \quad (37)$$

Corollary 1. *Given theorem 4, we have*

$$\begin{aligned} \text{Var} \left(\frac{C_1}{\text{tr}(\Sigma)} \right) &= O \left(\sum_{i=1}^g \frac{1}{n_i} \right) = \text{Var} \left(\frac{C_2}{[\text{tr}(\Sigma)]^2} \right) \\ \text{Var} \left(\frac{C_3}{\text{tr}(\Sigma^2)} \right) &= O \left(\sum_{i=1}^g \sum_{j=1}^g \frac{1}{n_i n_j} \right) \\ \text{Cov} \left(\frac{C_1}{\text{tr}(\Sigma)}, \frac{C_2}{[\text{tr}(\Sigma)]^2} \right) &= O \left(\sum_{i=1}^g \frac{1}{n_i} \right) \\ \text{Cov} \left(\frac{C_1}{\text{tr}(\Sigma)}, \frac{C_3}{\text{tr}(\Sigma^2)} \right) &= O \left(\sum_{i=1}^g \frac{1}{n_i^2} \right) = \text{Cov} \left(\frac{C_2}{[\text{tr}(\Sigma)]^2}, \frac{C_3}{\text{tr}(\Sigma^2)} \right). \end{aligned}$$

From equations (22) and (23), $\text{Var}(T_a) = \text{Var}(C_3/\text{tr}(\Sigma^2)) = \text{Var}(T_b)$, which, under H_{0a} or H_{0b} , reduce to $4/P^*$, by assumption 1. The following is then a direct extension of theorem 3 for $g \geq 2$; the proof of which also follows similarly.

Theorem 5. *Let T_a and T_b be as defined in equations (22) and (23), respectively. Then, under model (25) and assumptions 1 and 2, as $n_i, p \rightarrow \infty$,*

$$\begin{aligned} \sigma_{T_a}^{-1} \left(\frac{T_a + 1}{\psi} - 1 \right) &\xrightarrow{\mathcal{D}} N(0, 1) \\ \sigma_{T_b}^{-1} \left[T_b - \frac{1}{p} \text{tr}(\Sigma - \mathbf{I})^2 \right] &\xrightarrow{\mathcal{D}} N(0, 1), \end{aligned}$$

where $\sigma_{T_a}^2 = \sigma_{T_b}^2$ denote the variances of T_a and T_b . In particular, under H_{0j} and assumption 1, $\sqrt{P^*/4T_j} \xrightarrow{\mathcal{D}} N(0, 1)$, $j = a$ or $j = b$.

4. The non-normal case

This section evaluates the proposed tests by relaxing the normality assumption. Let $\mathbf{X}_{ik} \sim \mathcal{F}_i$ with $\mathbf{a}_{it} = \mathbf{X}_{it} - \boldsymbol{\mu}_{it}$, $i = 1, \dots, g$. We replace the normal model with the following general multivariate model

$$\mathbf{a}_{ik} = \mathbf{\Lambda}_i \mathbf{b}_{ik}, \quad (38)$$

with $\mathbf{b}_{ik} = (b_{ik1}, \dots, b_{ikp})'$ independent vectors, $E(\mathbf{b}_{ik}) = \mathbf{0}$, $\text{Cov}(\mathbf{b}_{ik}) = \mathbf{I}$, and $\mathbf{\Lambda}_i$ a known $p \times p$ matrix of constants such that $\mathbf{\Lambda}_i' \mathbf{\Lambda}_i = \mathbf{A}_i$ and $\mathbf{\Lambda}_i \mathbf{\Lambda}_i' = \Sigma_i > 0$. Model (38) covers a wide variety of non-normal multivariate distributions, some of which are used in

Section 5 to show numerical accuracy of the proposed tests and comparison with the normal case.

An important motivation of this extension is that most of the proofs under normality easily extend for model (38). In particular, the proofs of main theorems, 3 and 5, do not rest on normality assumption and hence remain valid under model (38). Similarly, with only a slight modification, the proof of theorem 2 fits under the general setup with the same asymptotic limits.

Note that model (38) is defined in mean-deviated form so that the location-invariance property of Section 3 remains intact. Thus, model (38) not only extends the scope of the proposed tests to a wider class of multivariate distributions, but it also coherently combines the normal and non-normal cases within the same framework. This also applies to the theoretical results for model (38), which reduce to their normal counterparts if $\mathcal{F}_i$ is multivariate normal.

For simplicity, we only focus on the multi-sample case, avoiding unnecessary details. First, we need the following additional assumptions under model (38).

Assumption 4. $E(b_{iks}^4) = 3 + \gamma > 0, \forall s = 1, \dots, p$, where $\gamma \in \mathbb{R}$.

Assumption 5. For $p \rightarrow \infty$, $\frac{\text{tr}(\mathbf{\Sigma}_i^i \odot \mathbf{\Sigma}_j^j)}{\text{tr}(\mathbf{\Sigma}_i^i \otimes \mathbf{\Sigma}_j^j)} \rightarrow 0, i, j = 1, 2, 3, i + j \leq 4$, where $\odot$ and $\otimes$ denote the Hadamard and Kronecker products, respectively.

Assumption 4 is a natural replacement of normality because the computations involve moments of quadratic and bilinear forms. Further, like assumptions under normality, assumption 5 is also mild and conveniently justifiable because the denominator in the ratio of traces is much heavier in p than the numerator.

Theorem 6. Let B_i and B_{ij} be the estimators of $\text{tr}(\mathbf{\Sigma}_i)$ and $\text{tr}(\mathbf{\Sigma}_i \mathbf{\Sigma}_j)$. Then, under model (38),

$$E(B_i) = \text{tr}(\mathbf{\Sigma}_i), \quad E(B_{12}) = \text{tr}(\mathbf{\Sigma}_1 \mathbf{\Sigma}_2) \quad (39)$$

$$\text{Var}(B_i) = \frac{2}{n_i - 1} \left[\text{tr}(\mathbf{\Sigma}^2) + M_1 O(1) \right] \quad (40)$$

$$\begin{aligned} \text{Var}(B_{ij}) = \frac{2}{(n_i - 1)(n_j - 1)} & \left[(n - 1) \text{tr}\{(\mathbf{\Sigma}_i \mathbf{\Sigma}_j)^2\} + [\text{tr}(\mathbf{\Sigma}_i \mathbf{\Sigma}_j)]^2 \right. \\ & \left. + M_2 O(n_{ij}) + M_3 O(1) \right] \end{aligned} \quad (41)$$

$$\text{Cov}(B_i, B_{ij}) = \frac{2}{n_i - 1} \left[\text{tr}(\mathbf{\Sigma}_i^2 \mathbf{\Sigma}_j) + M_2 O(1) \right], \quad i \neq j, \quad (42)$$

where $n_{ij} = n_i + n_j, i = 1, \dots, g$ and $M_1, M_2, \dots$ are given in theorem 11.

From theorem 11 and lemma 2, the terms $M_1, M_2, \dots$ involve the numerator of the trace ratio in assumption 5. Then, for estimators appropriately scaled by the traces they estimate (as used in T_a and T_b), it immediately follows from Assumption 5 that all ratios involving M_1, M_2 and others vanish as $n_i, p \rightarrow \infty$. This implies that the moment bounds of estimators in theorem 1 remain exactly the same under this assumption. By the same arguments, the proof of theorem 2 also follows with minimal adjustments. We state the general case of the theorem in the following with a brief sketch of its proof in Appendix B5.

Theorem 7. Theorem 2 remains valid for model (38) under assumptions 1, 4 and 5.

Using these results, we can now extend theorem 4 as follows.

Theorem 8. Let C_1 , C_2 and C_3 be as defined in equations (28)–(30). Then, under model (38), with $n_{ij} = n_i + n_j$,

$$E(C_1) = \text{tr}(\Sigma), \quad E(C_2) = [\text{tr}(\Sigma)]^2, \quad E(C_3) = \text{tr}(\Sigma^2) \quad (43)$$

$$\text{Var}(C_1) = \frac{2}{[\sum_{i=1}^g Q(n_i)]^2} \sum_{i=1}^g \left\{ n_i Q(n_i) \text{tr}(\Sigma^2) + M_1 O(n_i) \right\} \quad (44)$$

$$\begin{aligned} \text{Var}(C_2) = & \frac{8}{g^2(g-1)^2} \left(\sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g \frac{4}{P(n_i, n_j)} \left\{ n_i \text{tr}(\Sigma^2) + M_1 O(n_i) \right\} \{j\} \right. \\ & \left. + (g-1)^2 [\text{tr}(\Sigma)]^2 \sum_{i=1}^g \frac{1}{Q(n_i)} \left\{ n_i \text{tr}(\Sigma^2) + M_1 O(n_i) \right\} \right) \end{aligned} \quad (45)$$

$$\begin{aligned} \text{Var}(C_3) = & \frac{4}{P^{*2}} \left(\sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g P(n_i, n_j) \left[n_i n_j \left\{ (n_{ij} - 1) \text{tr}(\Sigma^4) + [\text{tr}(\Sigma^2)]^2 \right\} \right. \right. \\ & \left. \left. + M_2 O(1) \right] + g^* \left[32 \text{tr}(\Sigma^4) + 15 [\text{tr}(\Sigma^2)]^2 + M_2 O(1) \right] \right) \end{aligned} \quad (46)$$

$$\text{Cov}(C_1, C_2) = \frac{4 \sum_{i=1}^g n_i}{g \sum_{i=1}^g Q(n_i)} \text{tr}(\Sigma) \left\{ n_i \text{tr}(\Sigma^2) + M_1 O(n_i) \right\} \quad (47)$$

$$\text{Cov}(C_1, C_3) = \frac{4}{\sum_{i=1}^g Q(n_i)} \left\{ \text{tr}(\Sigma^3) + M_2 O(1) \right\} \quad (48)$$

$$\text{Cov}(C_2, C_3) = \frac{8}{P^*} \left(\text{tr}(\Sigma^4) + \text{tr}(\Sigma^3) \text{tr}(\Sigma) + \text{tr}(\Sigma^3) \text{tr}(\Sigma) \sum_{i=1}^g Q(n_i) \right. \quad (49)$$

$$\left. + \text{tr}(\Sigma) M_3 O(1) + M_4 O(1) \right) \quad (50)$$

where $\{j\} = n_j \text{tr}(\Sigma^2) + M_1 O(n_j)$ and $M_1, M_2, \dots$ are given in theorem 11.

By the same arguments as for theorem 6, the terms involving $M_1, \dots$ disappear so that the bounds in corollary 1 hold under assumption 5, as stated in the following.

Corollary 2. Given theorem 8. Under assumptions 1, 4 and 5, the bounds given in corollary 1 remain valid for model (38).

Using equations (22) and (23) and the same strategy as under normality, the following is a logical extension of theorem 5. The proof under the alternative follows from theorem 8 and corollary 2; under the null, the proof under normality holds as no normality assumption is used there.

Theorem 9. *Theorem 5 remains valid for model (38) under assumptions 1, 4 and 5.*

In Section 5, we use simulation results to demonstrate the accuracy of the proposed tests under the general model, thereby showing that, only at the cost of an extra mild assumption, the scope of the proposed tests is enhanced to a very wide class of multivariate models with discernable accuracy.

5. Simulations

As the results for both tests, and for two or more populations, are similar, we only present results of T_a for $g = 2$. Our aim is to show the accuracy of tests for small sample sizes and arbitrary dimension and also its robustness to normality. For the first objective, we generate random samples with $(n_1, n_2) = (5, 10), (10, 15), (5, 50)$; the last pair represents seriously unbalanced design ($n_2 = 10n_1$). Each pair of sample sizes is combined with $p = \{50, 100, 300, 500, 1000\}$. For the second objective, random samples are generated from four multivariate distributions: normal, exponential, uniform and t_ν ($\nu = 15$).

The test size is evaluated for $\alpha = \{0.01, 0.05, 0.10\}$, and the estimated tail probabilities $(1 - \alpha)$ of the tests are presented (Table 1). For power (Table 2), we use $\alpha = 0.05$ and two covariance structures under the alternative, compound symmetry (CS) and autoregressive of order 1, AR(1), defined respectively as $\Sigma = \kappa \mathbf{I} + \rho \mathbf{J}$ and $\text{Cov}(X_k, X_l) = \kappa \rho^{|k-l|}$, $\forall k, l$, with $\mathbf{I}$ as identity matrix and $\mathbf{J}$ as the matrix of ones, where κ and ρ are constants. We use $\kappa = 1$ and $\rho = 0.5$ for both CS and AR(1). All results are averages over 1000 simulations.

We observe that T_a accurately estimates test size and has high power for n_i as small as 5 or 10 and for any p , with the accuracy intact for all distributions, also for unequal sample sizes. Although, the expected heavy tailedness of t is apparent for small p , the result quickly converges with increasing p .

Note the difference of power across covariance structures and distributions. Whereas the power for CS is not only high but also similar, that under AR(1) is relatively low in the beginning but improves quickly with increasing p and particularly with moderate n_i . Note that the power under AR(1) is also investigated for other ρ values, for example, 0.3 or 0.7. It is observed that the power decreases, logically towards α , as ρ decreases, but goes rapidly up for increasing ρ ; for example, for $\rho = 0.7$, it is similar to that under CS. Likewise, t distribution with other degrees of freedom, for example, $\nu = 10$, produced similar results as reported earlier.

The difference of power between exponential distribution and other non-normal models is also obvious. Higher power for uniform and t under AR(1) apparently pertains to slightly overestimated tail probabilities observed in Table 1 corresponding to the same dimensions. In general, it can be concluded that the proposed tests not only attain the nominal test size accurately but also have high power for very small sample sizes and these properties are not disturbed even if the data follow a non-normal model or an unbalanced design.

6. Discussion and conclusions

Test statistics for different structures of a covariance matrix, assumed to be common across several multivariate populations, are presented when the dimension of the multivariate vector

Table 1. *Estimated $1 - \alpha$ of T_a for $g = 2$*

n_1	n_2	$1 - \alpha$	p				
			50	100	300	500	1000
Normal							
5	10	0.90	0.895	0.902	0.900	0.893	0.897
		0.95	0.940	0.939	0.942	0.940	0.942
		0.99	0.979	0.977	0.981	0.979	0.983
10	15	0.90	0.894	0.902	0.893	0.895	0.890
		0.95	0.943	0.942	0.940	0.943	0.944
		0.99	0.984	0.985	0.982	0.985	0.983
5	50	0.90	0.926	0.900	0.893	0.902	0.910
		0.95	0.959	0.945	0.955	0.949	0.956
		0.99	0.991	0.989	0.990	0.989	0.989
Exponential							
5	10	0.90	0.912	0.911	0.905	0.902	0.890
		0.95	0.959	0.956	0.943	0.947	0.940
		0.99	0.995	0.991	0.981	0.984	0.980
10	15	0.90	0.932	0.904	0.891	0.881	0.889
		0.95	0.968	0.966	0.953	0.945	0.946
		0.99	0.994	0.993	0.989	0.986	0.989
5	50	0.90	0.9540	0.939	0.905	0.895	0.899
		0.95	0.982	0.979	0.951	0.947	0.956
		0.99	1.000	0.995	0.990	0.992	0.991
Uniform							
5	10	0.90	0.906	0.913	0.903	0.891	0.890
		0.95	0.954	0.951	0.946	0.940	0.945
		0.99	0.984	0.987	0.983	0.991	0.985
10	15	0.90	0.923	0.918	0.895	0.906	0.898
		0.95	0.960	0.954	0.945	0.940	0.955
		0.99	0.991	0.991	0.990	0.980	0.992
5	50	0.90	0.917	0.897	0.905	0.902	0.898
		0.95	0.955	0.956	0.948	0.953	0.944
		0.99	0.991	0.989	0.991	0.992	0.988
t							
5	10	0.90	0.925	0.906	0.896	0.908	0.903
		0.95	0.968	0.963	0.947	0.940	0.950
		0.99	0.990	0.995	0.981	0.983	0.989
10	15	0.90	0.932	0.929	0.888	0.901	0.899
		0.95	0.972	0.964	0.945	0.951	0.948
		0.99	0.994	0.987	0.988	0.986	0.993
5	50	0.90	0.950	0.938	0.903	0.918	0.891
		0.95	0.978	0.965	0.953	0.955	0.940
		0.99	1.000	0.990	0.991	0.993	0.980

Table 2. Estimated $1 - \beta$ of T_a for $g = 2$

Σ	n_1	n_2	p				
			50	100	300	500	1000
Normal							
CS	5	10	0.973	0.993	0.999	1.000	1.000
	10	15	0.999	1.000	1.000	1.000	1.000
	5	50	0.991	1.000	1.000	1.000	1.000
AR(1)	5	10	0.609	0.685	0.724	0.730	0.741
	10	15	0.897	0.947	0.983	0.987	0.989
	5	50	0.899	0.972	0.992	0.997	1.000
Exponential							
CS	5	10	0.873	0.960	0.992	1.000	1.000
	10	15	0.988	0.997	1.000	1.000	1.000
	5	50	0.937	0.985	0.997	1.000	1.000
AR(1)	5	10	0.454	0.500	0.586	0.602	0.622
	10	15	0.696	0.779	0.898	0.930	0.945
	5	50	0.679	0.808	0.940	0.965	0.987
Uniform							
CS	5	10	0.972	0.990	1.000	1.000	1.000
	10	15	0.998	1.000	1.000	1.000	1.000
	5	50	0.985	0.997	1.000	1.000	1.000
AR(1)	5	10	0.625	0.697	0.710	0.721	0.744
	10	15	0.912	0.962	0.987	0.998	1.000
	5	50	0.917	0.972	0.996	1.000	1.000
t							
CS	5	10	0.983	0.997	1.000	1.000	1.000
	10	15	0.995	1.000	1.000	1.000	1.000
	5	50	0.964	0.998	1.000	1.000	1.000
AR(1)	5	10	0.651	0.703	0.720	0.742	0.752
	10	15	0.918	0.972	0.990	1.000	1.000
	5	50	0.908	0.975	0.996	0.998	1.000

CS, compound symmetry; AR(1), autoregressive of order 1.

may be much larger than the number of independent vectors. The test statistics are location-invariant, and functions of estimators are defined as U -statistics, hence the name U -tests, which are non-degenerate under the alternative and degenerate under the null hypothesis. The distribution-free nature of U -statistics-based estimators are exploited to extend the proposed tests to a larger class of multivariate models. Theoretical properties of the tests and their asymptotic distributions are derived under minimal and mild assumptions. The high-dimensional small-sample accuracy of the tests is also shown through simulation studies covering both normal and non-normal models.

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Appendix A: Basic results

A1. U -statistics

We briefly summarize basic results of U -statistics to be used later. For details, see, e.g., Lehmann (1999, Ch. 6), Koroljuk & Borovskich (1994), (Sering, 1980, Ch. 5). For a symmetric function (kernel), $h(X_1, \dots, X_m) : \mathcal{R}^m \rightarrow \mathcal{R}$ with X_i 's iid, an m th order U -statistic, U_n , with $E(U_n) = \theta = E[h(\cdot)]$, is defined as

$$U_n = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} h(X_{i_1}, \dots, X_{i_m}), \quad (51)$$

with summation over all subsets of m distinct elements taken from $\{1, \dots, n\}$. The projection $h_c(x_1, \dots, x_c) = E(h(X_1, \dots, X_m) | X_1 = x_1, \dots, X_c = x_c)$ with $h_m(\cdot) = h(\cdot)$ and $\xi_c = \text{Var}(h_c(x_1, \dots, x_c))$, $c = 1, \dots, m$, gives

$$\text{Var}(U_n) = \binom{n}{m}^{-1} \sum_{c=1}^m \binom{m}{c} \binom{n-m}{m-c} \xi_c. \quad (52)$$

If $0 < \xi_1 < \infty$ and $\xi_c < \infty \forall c = 1, \dots, m$, then (Lehmann, 1999, p. 369)

$$\frac{U_n - E(U_n)}{\sqrt{\text{Var}(U_n)}} \xrightarrow{D} N(0, 1). \quad (53)$$

For two U -statistics, $U_n^{(1)}, U_n^{(2)}$ from same sample with kernels $h_i(X_1, \dots, X_{m_i})$, projections $h_{ic}(\cdot)$, $i = 1, 2$, let $\xi_{cc} = \text{Cov}(h_{1,c}(X_1, \dots, X_c), h_{2,c}(X_1, \dots, X_c))$ if they share c elements. Then, for $m_1 \leq m_2$,

$$\text{Cov}(U_n^{(1)}, U_n^{(2)}) = \binom{n}{m_1}^{-1} \sum_{c=1}^{m_1} \binom{m_2}{c} \binom{n-m_2}{m_1-c} \xi_{cc}. \quad (54)$$

As another extension, consider two-sample U -statistic

$$U_{n_1 n_2} = \left[\binom{n_1}{m_1} \binom{n_2}{m_2} \right]^{-1} \sum_{P(i)} \sum_{P(j)} h(X_{i_1}, \dots, X_{i_{m_1}}; X_{j_1}, \dots, X_{j_{m_2}}), \quad (55)$$

where $P(i) = 1 \leq i_1 < \dots < i_{m_1} \leq n_1$ and $P(j) = 1 \leq j_1 < \dots < j_{m_2} \leq n_2$ cover all possible subsets of m_i distinct elements taken from $\{1, \dots, n_i\}$, $i = 1, 2$ and $h(\cdot)$ is symmetric in elements within each sample. The projection $h_{c_1 c_2} = E[h(\cdot) | X_1, \dots, X_{c_1}; X_1, \dots, X_{c_2}]$, gives $\xi_{c_1 c_2} = \text{Cov}[h(\cdot), h_{c_1 c_2}(\cdot)]$, with $\xi_{00} = 0$, $c_1 = 0, 1, \dots, m_1$, $c_2 = 0, 1, \dots, m_2$. Then, (Lee, 1990, p. 38)

$$\text{Var}(U_{n_1 n_2}) = \left[\binom{n_1}{m_1} \binom{n_2}{m_2} \right]^{-1} \sum_{c_1=0}^{m_1} \sum_{c_2=0}^{m_2} \binom{m_1}{c_1} \binom{n_1-m_1}{m_1-c_1} \binom{m_2}{c_2} \binom{n_2-m_2}{m_2-c_2} \xi_{c_1 c_2}. \quad (56)$$

Further, if $0 \leq n_i/N \leq 1$, $N = n_1 + n_2$, $0 < \xi_{01}, \xi_{10} < \infty$, and $\xi_{c_1 c_2} < \infty$, $\forall c_i = 1, \dots, m_i$, $i = 1, 2$, then (Lehmann, 1999, Theorem 6.1.4(ii), p. 375)

$$\frac{U_{n_1 n_2} - E(U_{n_1 n_2})}{\sqrt{\text{Var}(U_{n_1 n_2})}} \xrightarrow{D} N(0, 1). \quad (57)$$

A2. Quadratic and bilinear forms

Theorem 10 collects basic moments of quadratic and bilinear forms, frequently available in the literature. On the basis of these results, we prove in lemma 1 several extended results to be used in the main proofs. The proofs of these results, tedious but not directly relevant here, are skipped; see Ahmad (2014b).

Theorem 10. (Searle, 1971; Mathai *et al.*, 1995) Let $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \Sigma)$, $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \Sigma)$, $\Sigma > 0$, be $p \times 1$ independent random vectors, and $\mathbf{A}, \mathbf{B}$ $p \times p$ symmetric matrices. Let $Q_1 = \mathbf{u}'\mathbf{A}\mathbf{u}$, $Q_2 = \mathbf{u}'\mathbf{B}\mathbf{u}$, $B = \mathbf{u}'\mathbf{A}\mathbf{v}$. Then,

$$E(Q_1) = \text{tr}(\mathbf{A}\Sigma) \quad (58)$$

$$E(Q_1^2) = 2\text{tr}(\mathbf{A}\mathbf{\Sigma})^2 + [\text{tr}(\mathbf{A}\mathbf{\Sigma})]^2 \quad (59)$$

$$E(B) = 0 = E(B^3) \quad (60)$$

$$E(B^2) = \text{tr}(\mathbf{A}\mathbf{\Sigma})^2 \quad (61)$$

$$E(B^4) = 6\text{tr}(\mathbf{A}\mathbf{\Sigma})^2 + 3[\text{tr}(\mathbf{A}\mathbf{\Sigma})]^2 \quad (62)$$

$$\text{Cov}(Q_1, B) = 0 \quad (63)$$

$$\text{Cov}(Q_1, Q_2) = 2\text{tr}(\mathbf{A}\mathbf{\Sigma}\mathbf{B}\mathbf{\Sigma}) \quad (64)$$

$$\text{Cov}(Q_1^2, Q_2) = 4\text{tr}(\mathbf{A}\mathbf{\Sigma})\text{tr}(\mathbf{A}\mathbf{\Sigma}\mathbf{B}\mathbf{\Sigma}) + 8\text{tr}[(\mathbf{A}\mathbf{\Sigma})^2\mathbf{B}\mathbf{\Sigma}] \quad (65)$$

$$\begin{aligned} \text{Var}(Q_1 Q_2) = & 32\text{tr}[(\mathbf{A}\mathbf{\Sigma})^2(\mathbf{B}\mathbf{\Sigma})^2] + 16\left[\text{tr}(\mathbf{A}\mathbf{\Sigma}\mathbf{B}\mathbf{\Sigma})^2 \right. \\ & + \text{tr}(\mathbf{A}\mathbf{\Sigma})\text{tr}\{(\mathbf{A}\mathbf{\Sigma})(\mathbf{B}\mathbf{\Sigma})^2\} + \text{tr}(\mathbf{B}\mathbf{\Sigma})\text{tr}\{(\mathbf{A}\mathbf{\Sigma})^2(\mathbf{B}\mathbf{\Sigma})\}] \\ & + 4\left[\text{tr}(\mathbf{A}\mathbf{\Sigma})^2\text{tr}(\mathbf{B}\mathbf{\Sigma})^2 + [\text{tr}(\mathbf{A}\mathbf{\Sigma}\mathbf{B}\mathbf{\Sigma})]^2 + \text{tr}(\mathbf{A}\mathbf{\Sigma})\text{tr}(\mathbf{B}\mathbf{\Sigma})\text{tr}(\mathbf{A}\mathbf{\Sigma}\mathbf{B}\mathbf{\Sigma})\right] \\ & + 2\left[[\text{tr}(\mathbf{A}\mathbf{\Sigma})]^2\text{tr}(\mathbf{B}\mathbf{\Sigma})^2 + [\text{tr}(\mathbf{B}\mathbf{\Sigma})]^2\text{tr}(\mathbf{A}\mathbf{\Sigma})^2\right] \end{aligned} \quad (66)$$

Lemma 1. Let $\mathbf{a}_{it} = \mathbf{X}_{it} - \boldsymbol{\mu}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_i)$ be independent vectors. Then,

$$E[\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{iu}\boldsymbol{\Sigma}_i\mathbf{a}_{iv}] = \text{tr}(\boldsymbol{\Sigma}_i^4) \quad (67)$$

$$E[\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{iw}\mathbf{a}_{iu}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{iw}\mathbf{a}_{iv}] = \text{tr}(\boldsymbol{\Sigma}_i^4) \quad (68)$$

$$E[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\boldsymbol{\Sigma}_i\mathbf{a}_{it}] = 2\text{tr}(\boldsymbol{\Sigma}_i^4) + [\text{tr}(\boldsymbol{\Sigma}_i^2)]^2 \quad (69)$$

$$\text{Var}(\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{iv}\mathbf{a}_{iu}) = 2\text{tr}(\boldsymbol{\Sigma}_i^4) + [\text{tr}(\boldsymbol{\Sigma}_i^2)]^2 \quad (70)$$

$$\text{Cov}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2, (\mathbf{a}'_{it}\mathbf{a}_{iv})^2] = 2\text{tr}(\boldsymbol{\Sigma}_i^4) \quad (71)$$

$$E[(\mathbf{a}'_{it}\mathbf{a}_{ju})^2\mathbf{a}'_{it}\boldsymbol{\Sigma}_j\mathbf{a}_{it}] = 2\text{tr}\{(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2\} + [\text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)]^2 \quad (72)$$

$$\text{Var}(\mathbf{a}'_{it}\mathbf{a}_{ju}\mathbf{a}'_{iv}\mathbf{a}_{ju}) = 2\text{tr}\{(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2\} + [\text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)]^2 \quad (73)$$

$$\text{Cov}[(\mathbf{a}'_{jt}\mathbf{a}_{iu})^2, (\mathbf{a}'_{jt}\mathbf{a}_{iv})^2] = 2\text{tr}\{(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2\} \quad (74)$$

$$\text{E}(\mathbf{a}'_{jt}\mathbf{a}_{iu}\mathbf{a}'_{jt}\mathbf{a}_{iv}\mathbf{a}'_{iu}\boldsymbol{\Sigma}_j\mathbf{a}_{iv}) = \text{tr}\{(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2\} \quad (75)$$

$$\text{Cov}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2, \mathbf{a}'_{it}\boldsymbol{\Sigma}_j\mathbf{a}_{it}] = 2\text{tr}(\boldsymbol{\Sigma}_i^3\boldsymbol{\Sigma}_j) \quad (76)$$

$$\text{Cov}(\mathbf{a}'_{it}\boldsymbol{\Sigma}_i\mathbf{a}_{iu}, \mathbf{a}'_{it}\boldsymbol{\Sigma}_j\mathbf{a}_{iu}) = \text{tr}\{(\boldsymbol{\Sigma}_i^3\boldsymbol{\Sigma}_j)\} \quad (77)$$

$$\text{E}[(\mathbf{a}'_{iu}\mathbf{a}_{iv})^2\mathbf{a}'_{it}\boldsymbol{\Sigma}_j\mathbf{a}_{it}] = \text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)\text{tr}(\boldsymbol{\Sigma}_i)^2, \quad (78)$$

where $\text{E}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{it}\mathbf{a}_{iv}]$, $\text{E}[\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{it}\boldsymbol{\Sigma}_i\mathbf{a}_{it}]$, $\text{E}[\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{it}\boldsymbol{\Sigma}_i\mathbf{a}_{iu}]$, $\text{E}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\boldsymbol{\Sigma}_i\mathbf{a}_{iu}]$, $\text{E}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\boldsymbol{\Sigma}_j\mathbf{a}_{iu}]$, $\text{E}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{iu}\mathbf{a}_{iv}]$ and $\text{E}[\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{iu}\mathbf{a}_{iv}]$ all vanish.

Some moments in theorem 10 and lemma 1 change for model (38). For $\mathbf{a}_{ik}$ as in Section 4, let $A_{ik} = \mathbf{a}'_{ik}\boldsymbol{\Sigma}_i\mathbf{a}_{ik} = \mathbf{b}'_{ik}\mathbf{A}^2\mathbf{b}_{ik}$, and $A_{ijkl} = \mathbf{a}'_{ik}\mathbf{a}_{jl} = \mathbf{b}'_{ik}\mathbf{A}_i\mathbf{A}_j\mathbf{b}_{jl}$, $k \neq l$, be quadratic and bilinear forms, with $A_{ik} = \mathbf{b}_{ik}\mathbf{A}_i\mathbf{b}_{ik} = Q_{ik}$ for $\boldsymbol{\Sigma}_i = \mathbf{I}$. As all terms involving $\mathbf{A}_i$ will eventually vanish under assumption 5, we only write $\mathbf{A}_i = \mathbf{A}$ for simplicity and without loss of generality. Theorem 11 and lemma 2 summarize the moments.

Theorem 11. For A_{ik} and A_{ijkl} , as defined earlier, we have

$$\text{E}\left(Q_{ik}^2\right)^2 = 2\text{tr}(\boldsymbol{\Sigma}_i^2) + [\text{tr}(\boldsymbol{\Sigma}_i)]^2 + M_1 \quad (79)$$

$$\text{E}\left(A_{ik}^2\right)^2 = 2\text{tr}(\boldsymbol{\Sigma}_i^4) + [\text{tr}(\boldsymbol{\Sigma}_i^2)]^2 + M_2 \quad (80)$$

$$\text{E}(A_{ik}A_{jk}) = 2\text{tr}(\boldsymbol{\Sigma}_i^3\boldsymbol{\Sigma}_j) + \text{tr}(\boldsymbol{\Sigma}_i^2)\text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j) + M_2 \quad (81)$$

$$\text{E}\left(A_{ijkl}^4\right) = 6\text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2 + 3[\text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)]^2 + M_3 \quad (82)$$

$$\text{E}\left(Q_{ik}Q_{jk}A_{ijkl}^2\right) = 4\text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2 + 4\text{tr}(\boldsymbol{\Sigma}_i^3)\text{tr}(\boldsymbol{\Sigma}_j) + [\text{tr}(\boldsymbol{\Sigma}_i)]^2\text{tr}(\boldsymbol{\Sigma}_j^2) + M_4 \quad (83)$$

with $M_1 = \gamma\text{tr}(\mathbf{A} \odot \mathbf{A})$, $M_2 = \gamma\text{tr}(\mathbf{A}^2 \odot \mathbf{A}^2)$, $M_3 = 6\gamma\text{tr}(\mathbf{A}^2 \odot \mathbf{A}^2) + \gamma^2 \sum_{s=1}^p \sum_{t=1}^p A_{st}^4$, $M_4 = 2\gamma\text{tr}(\boldsymbol{\Sigma}_i)\text{tr}(\mathbf{A}^2 \odot \mathbf{A}) + 4\gamma\text{tr}(\mathbf{A}^3 \odot \mathbf{A}) + \gamma\text{tr}(\mathbf{A} \odot \mathbf{A}\mathbf{D}\mathbf{A})$ and $\mathbf{D} = \text{diag}(\mathbf{A})$. Note also that, for a quadratic form, $A_{ik} = \mathbf{a}'_{ik}\mathbf{A}_i\mathbf{a}_{ik}$ and a bilinear form $A_{ikr} = \mathbf{a}'_{ik}\mathbf{a}_{ir}$, with $\mathbf{a}_{ik}$, $\mathbf{a}_{ir}$ independent, it trivially follows that $\text{E}(A_{ik}) = \text{tr}(\mathbf{A}_i\boldsymbol{\Sigma}_i)$, $\text{E}(A_{ikr}^2) = \text{tr}(\boldsymbol{\Sigma}_i^2)$, and $\text{Cov}(A_{ik}, A_{ikr}) = 0$, without normality assumption.

Lemma 2. Given $\mathbf{a}_{it}$ as defined earlier and M_2 as given in theorem 11, then the following moments of lemma 1 change under model (38), where all other moments remain the same.

$$E[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\boldsymbol{\Sigma}_i\mathbf{a}_{it}] = 2\text{tr}(\boldsymbol{\Sigma}_i^4) + [\text{tr}(\boldsymbol{\Sigma}_i^2)]^2 + M_2 \quad (84)$$

$$\text{Var}(\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{iv}\mathbf{a}_{iu}) = 2\text{tr}(\boldsymbol{\Sigma}_i^4) + [\text{tr}(\boldsymbol{\Sigma}_i^2)]^2 + M_2 \quad (85)$$

$$\text{Cov}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2, (\mathbf{a}'_{it}\mathbf{a}_{iv})^2] = 2\text{tr}(\boldsymbol{\Sigma}_i^4) + M_2 \quad (86)$$

$$E[(\mathbf{a}'_{it}\mathbf{a}_{ju})^2\mathbf{a}'_{it}\boldsymbol{\Sigma}_j\mathbf{a}_{it}] = 2\text{tr}\{(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2\} + [\text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)]^2 + M_2 \quad (87)$$

$$\text{Var}(\mathbf{a}'_{it}\mathbf{a}_{ju}\mathbf{a}'_{iv}\mathbf{a}_{ju}) = 2\text{tr}\{(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2\} + [\text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)]^2 + M_2 \quad (88)$$

$$\text{Cov}[(\mathbf{a}'_{jt}\mathbf{a}_{iu})^2, (\mathbf{a}'_{jt}\mathbf{a}_{iv})^2] = 2\text{tr}\{(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2\} + M_2 \quad (89)$$

$$\text{Cov}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2, \mathbf{a}'_{it}\boldsymbol{\Sigma}_j\mathbf{a}_{it}] = 2\text{tr}(\boldsymbol{\Sigma}_i^3\boldsymbol{\Sigma}_j) + M_2 \quad (90)$$

Appendix B: Main proofs

B1. Proof of theorem 1

Unbiasedness is trivial. For variances, consider B_i , a one-sample U -statistic with kernel (Appendix A1) $h(\mathbf{X}_{ik}, \mathbf{X}_{ir}) = A_{ikr} = (\mathbf{a}'_{ik}\mathbf{a}_{ik} - 2\mathbf{a}'_{ik}\mathbf{a}_{ir} + \mathbf{a}'_{ir}\mathbf{a}_{ir})/2$. Then,

$$h_1(\mathbf{X}_{ik}) = (\mathbf{a}'_{ik}\mathbf{a}_{ik} + \text{tr}(\boldsymbol{\Sigma}^2))/2 \Rightarrow \xi_1 = \text{tr}(\boldsymbol{\Sigma}_i^2)/2$$

$$h_2(\mathbf{X}_{ik}, \mathbf{X}_{ir}) = A_{ikr}/2 \Rightarrow \xi_2 = 2\text{tr}(\boldsymbol{\Sigma}_i^2),$$

using theorem 10. From equation (52)

$$\text{Var}(B_i) = \frac{2}{n_i(n_i - 1)} \left(2(n_i - 2) \cdot \frac{1}{2} \text{tr}(\boldsymbol{\Sigma}_i^2) + 2\text{tr}(\boldsymbol{\Sigma}_i^2) \right) = \frac{2}{n_i - 1} \text{tr}(\boldsymbol{\Sigma}_i^2).$$

For $\text{Var}(B_{ij})$, $h(\mathbf{X}_{ik}, \mathbf{X}_{ir}, \mathbf{X}_{jl}, \mathbf{X}_{js}) = A_{ijkrls}^2/4$. We use i and j for the sake of generality. Writing $B_{ij} = U_{n_in_j} = \frac{1}{4}U$ and ignoring 4 for simplicity,

$$A_{ijkrls}^2 = h(\cdot) = [(\mathbf{a}_{ik} - \mathbf{a}_{ir})'(\mathbf{a}_{jl} - \mathbf{a}_{js})]^2 = [\mathbf{a}'_{ik}\mathbf{a}_{jl} - \mathbf{a}'_{ik}\mathbf{a}_{js} - \mathbf{a}'_{ir}\mathbf{a}_{jl} + \mathbf{a}'_{ir}\mathbf{a}_{js}]^2,$$

with $h_{c_ic_j}(\cdot) = E[h(\cdot)|\mathbf{x}_{ik}, \mathbf{x}_{jl}, \dots]$, $\xi_{c_ic_j} = \text{Var}(h_{c_ic_j})$, $c_i = 0, 1, 2$, $i = 1, 2$. Now,

$$h_{10} = E[h(\cdot)|\mathbf{x}_{ik}] = 2\mathbf{a}'_{ik}\boldsymbol{\Sigma}_j\mathbf{a}_{ik} + 2\text{tr}(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j) \Rightarrow \xi_{i0} = 8\text{tr}\{(\boldsymbol{\Sigma}_i\boldsymbol{\Sigma}_j)^2\},$$

with the same result for other components, by symmetry. Similarly,

$$h_{20} = E[h(\cdot)|\mathbf{x}_{ik}, \mathbf{x}_{ir}] = 2(\mathbf{a}_{ik} - \mathbf{a}_{ir})'\boldsymbol{\Sigma}_j(\mathbf{a}_{ik} - \mathbf{a}_{ir}),$$

so that $\xi_{20} = 32\{\text{tr}(\Sigma_i \Sigma_j)^2\}$, where $\xi_{20} = \xi_{02}$. Further,

$$\begin{aligned} h_{11} &= E[h(\cdot) | \mathbf{x}_{ik}, \mathbf{x}_{jl}] = (\mathbf{a}'_{ik} \mathbf{a}_{jl})^2 + \mathbf{a}'_{ik} \Sigma_j \mathbf{a}_{ik} + \mathbf{a}'_{jl} \Sigma_i \mathbf{a}_{jl} + \text{tr}(\Sigma_i \Sigma_j) \\ \xi_{11} &= 18\text{tr}\{(\Sigma_i \Sigma_j)^2\} + 2[\text{tr}(\Sigma_i \Sigma_j)]^2 \\ h_{21} &= E[h(\cdot) | \mathbf{x}_{ik}, \mathbf{x}_{ir}, \mathbf{x}_{jl}] = (\mathbf{a}_{ik} - \mathbf{a}_{ir})' \Sigma_j (\mathbf{a}_{ik} - \mathbf{a}_{ir}) + [(\mathbf{a}_{ik} - \mathbf{a}_{ir})' \mathbf{a}_{jl}]^2, \end{aligned}$$

with variances of the last two terms $8\text{tr}\{(\Sigma_i \Sigma_j)^2\}$ and $24\text{tr}\{(\Sigma_i \Sigma_j)^2\} + 8[\text{tr}(\Sigma_i \Sigma_j)]^2$, using theorem 10. The first part of their covariance, using trace property and $E(\mathbf{a}_{js} \mathbf{a}'_{js}) = \Sigma_j$, is $E[(\mathbf{a}_{ik} - \mathbf{a}_{ir})' \Sigma_j (\mathbf{a}_{ik} - \mathbf{a}_{ir})^2]$ so that, by theorem 10, the covariance reduces to $8\text{tr}\{(\Sigma_i \Sigma_j)^2\}$. This finally gives

$$\xi_{21} = 48\text{tr}\{(\Sigma_i \Sigma_j)^2\} + 8[\text{tr}(\Sigma_i \Sigma_j)]^2,$$

where it can be verified that $\xi_{12} = \xi_{21}$. Finally, $h_{22}(\cdot) = h(\cdot)$ so that

$$\xi_{22} = \text{Var}(A_{ijkl}^2) = 96\text{tr}\{(\Sigma_i \Sigma_j)^2\} + 32[\text{tr}(\Sigma_i \Sigma_j)]^2,$$

and equation (56) gives $\text{Var}(B_{ij})$. $\text{Cov}(B_1, B_2) = 0$, by independence. Using i, j for the sake of generality, $\text{Cov}(B_i, B_{ij}) = E(B_i B_{ij}) - \text{tr}(\Sigma_i^2) \text{tr}(\Sigma_i \Sigma_j)$ with

$$E(B_i B_{ij}) = \frac{1}{8Q(n_i)P(n_i, n_j)} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{\substack{r=1 \\ r \neq k}}^{n_i} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{\substack{r=1 \\ r \neq k}}^{n_i} \sum_{\substack{l=1 \\ l \neq s}}^{n_j} \sum_{\substack{s=1 \\ s \neq l}}^{n_j} E[A_{ikr} A_{ijkr}^2].$$

Writing $A'_{ikr} = \mathbf{D}'_{ikr} \Sigma_j \mathbf{D}_{ikr}$, we note that

$$E[A_{ikr} A_{ijkr}^2] = E[\text{tr}\{A_{ikr} \cdot \mathbf{D}'_{ikr} \mathbf{D}_{jls} \mathbf{D}'_{jls} \mathbf{D}_{ikr}\}] = 2E[A_{ikr} A'_{ikr}].$$

Adjusting $E(B_i B_{ij})$ accordingly, we can write it as a product of two U -statistics, each of second order but with kernels $A_{ikr}/2$ and $A'_{ikr}/2$, as follows.

$$\begin{aligned} E(B_i B_{ij}) &= \frac{1}{Q(n_i)Q(n_j)} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{\substack{r=1 \\ r \neq k}}^{n_i} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{\substack{r=1 \\ r \neq k}}^{n_i} \frac{1}{4} E[A_{ikr} A'_{ikr}] \\ &= E \left[\frac{1}{Q(n_i)} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{\substack{r=1 \\ r \neq k}}^{n_i} \frac{1}{2} A_{ikr} \cdot \frac{1}{Q(n_i)} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{\substack{r=1 \\ r \neq k}}^{n_i} \frac{1}{2} A'_{ikr} \right] \\ &= E(U_i^{(1)} U_i^{(2)}) = \text{Cov}(U_i^{(1)}, U_i^{(2)}) + E(U_i^{(1)}) E(U_i^{(2)}) \\ &= \text{Cov}(U_i^{(1)}, U_i^{(2)}) + \text{tr}(\Sigma_i^2) \text{tr}(\Sigma_i \Sigma_j), \end{aligned}$$

so that $\text{Cov}(B_i, B_{ij}) = \text{Cov}(U_i, U_j) = \text{Cov}(B'_i, B'_j)$ where B'_i is as B_i only with different kernel. From equation (54) with $m_1 = 2 = m_2$, we only need ξ_{11} and ξ_{22} . With $\mathbf{a}_{ik} = \mathbf{X}_{ik} - \boldsymbol{\mu}_i$, $\mathbf{D}'_{ikr} \Sigma_j \mathbf{D}_{ikr} = \mathbf{a}'_{ik} \Sigma_j \mathbf{a}_{ik} - 2\mathbf{a}'_{ik} \Sigma_j \mathbf{a}_{ir} + \mathbf{a}'_{ir} \Sigma_j \mathbf{a}_{ir}$, where taking h_{11}, h_{12} directly from $\text{Var}(B_i)$, we have

$$\begin{aligned}
h_{11} &= E[h_1(\mathbf{X}_{ik}, \mathbf{X}_{ir}) | \mathbf{x}_{ik}] = (\mathbf{a}'_{ik} \mathbf{a}_{ik} + \text{tr}(\boldsymbol{\Sigma}_i))/2 \\
h_{12} &= E[h_1(\mathbf{X}_{ik}, \mathbf{X}_{ir}) | \mathbf{x}_{ik}, \mathbf{x}_{ir}] = A_{ikr} \\
h_{21} &= E[h_2(\mathbf{X}_{ik}, \mathbf{X}_{ir}) | \mathbf{x}_{ik}] = (\mathbf{a}'_{ik} \boldsymbol{\Sigma}_j \mathbf{a}_{ik} + \text{tr}(\boldsymbol{\Sigma}_i \boldsymbol{\Sigma}_j))/2 \\
h_{22} &= E[h_2(\mathbf{X}_{ik}, \mathbf{X}_{ir}) | \mathbf{x}_{ik}, \mathbf{x}_{ir}] = A'_{ikr}.
\end{aligned}$$

Now, $\text{Cov}(B_i, B_{ij})$ follows from lemma 1 and equation (54) using

$$\begin{aligned}
\xi_{11} &= \text{Cov}(h_{11}, h_{21}) = \frac{1}{4} \text{Cov}(\mathbf{a}'_{ik} \mathbf{a}_{ik}, \mathbf{a}'_{ik} \boldsymbol{\Sigma}_j \mathbf{a}_{ik}) = \frac{1}{2} \text{tr}(\boldsymbol{\Sigma}_i^2 \boldsymbol{\Sigma}_j), \\
\xi_{22} &= \text{Cov}(h_{12}, h_{22}) = \frac{1}{4} [\text{Cov}(\mathbf{a}'_{ik} \mathbf{a}_{ik}, \mathbf{a}'_{ik} \boldsymbol{\Sigma}_j \mathbf{a}_{ik}) + \text{Cov}(\mathbf{a}'_{ir} \mathbf{a}_{ir}, \mathbf{a}'_{ir} \boldsymbol{\Sigma}_j \mathbf{a}_{ir}) \\
&\quad + 4 \text{Cov}(\mathbf{a}'_{ik} \mathbf{a}_{ir}, \mathbf{a}'_{ik} \boldsymbol{\Sigma}_j \mathbf{a}_{ir})] = 2 \text{tr}(\boldsymbol{\Sigma}_i^2 \boldsymbol{\Sigma}_j).
\end{aligned}$$

Finally, the bounds follow by the Cauchy–Schwarz inequality.

B2. Proof of theorem 2

The proof for both estimators follows the same strategy. We need to compute $\hat{U}_n$, the projection of (centred and appropriately scaled) U -statistic, say $\tilde{U}_n$, as a sum of iid random variables so that $\tilde{U}_n - \hat{U}_n = o_P(1)$ with $\text{Var}(\tilde{U}_n)/\text{Var}(\hat{U}_n) \rightarrow 1$. The asymptotic normality then follows from Lindeberg–Lévy central limit theorem (Polansky, 2011, p. 195) under the condition of square integrability. To allow for arbitrary p , the square-integrability condition needs to be carefully ensured. Note that the entire proof is based on first-order or Hajék projection of U -statistics, which implies that all limits will be non-degenerate with order $n^{1/2}$. Then, the bounds of the moment-ratios established in theorem 1, based on assumption 1, guarantees the required square-integrability condition, hence the non-degenerate limits.

For first-order projection of B_i with bivariate kernel, let $\tilde{U}_{n_i} = B_i/\text{tr}(\boldsymbol{\Sigma}_i) - 1$ with $\tilde{h}(\mathbf{X}_{ik}, \mathbf{X}_{ir}) = h(\mathbf{X}_{ik}, \mathbf{X}_{ir}) - 1$, $h(\cdot) = B_i/\text{tr}(\boldsymbol{\Sigma}_i)$, $E[h(\cdot)] = 1 = E(U_{n_i})$. From the proof of theorem 1, we have $\hat{U}_{n_i} = m \sum_{k=1}^{n_i} g(\mathbf{x}_{ik})/n_i$, with $m = 2$, $g(\mathbf{x}_{ik}) = h_1/\text{tr}(\boldsymbol{\Sigma}_i) - 1$ and $h_1(\cdot) = [\mathbf{a}'_{ik} \mathbf{a}_{ik} + \text{tr}(\boldsymbol{\Sigma}_i)]/2$ (Serfling 1980, p. 178). As $E[g(\mathbf{x}_{ik})] = 0 = E(\hat{U}_{n_i})$ and $\text{Var}[g(\mathbf{x}_{ik})] = \text{tr}(\boldsymbol{\Sigma}_i^2)/2[\text{tr}(\boldsymbol{\Sigma}_i)]^2$,

$$\text{Var}(\hat{U}_{n_i}) = \frac{2}{n_i} \cdot \frac{\text{tr}(\boldsymbol{\Sigma}_i^2)}{[\text{tr}(\boldsymbol{\Sigma}_i)]^2} = O\left(\frac{1}{n}\right),$$

uniformly bounded in p . Now, $\tilde{U}_{n_i} = \hat{U}_{n_i} + o_P(1)$ and $\text{Cov}(\hat{U}_{n_i}, \tilde{U}_{n_i}) = \text{Var}(\hat{U}_{n_i})$. As $\hat{U}_{n_i}$, a sum of iid random variables, is asymptotically normal by the central limit theorem, the same holds for $\tilde{U}_n$ by Slutsky's lemma. Proof of $\hat{B}_{ij} = B_{ij}/\text{tr}(\boldsymbol{\Sigma}_i \boldsymbol{\Sigma}_j) - 1$ is similar. With $\tilde{h}(\cdot) = A^2_{ijkrls}/\text{tr}(\boldsymbol{\Sigma}_i \boldsymbol{\Sigma}_j) - 1$, $g_{10} = \tilde{h}_{i0}$, $g_{01} = \tilde{h}_{01}$,

$$\begin{aligned}
\hat{U}_{n_i n_j} &= \frac{m_i}{n_i} \sum_{k=1}^{n_i} g_{i0}(\mathbf{x}_{ik}) + \frac{m_j}{n_j} \sum_{l=1}^{n_j} g_{01}(\mathbf{x}_{jl}) \\
&= \frac{1}{\text{tr}(\boldsymbol{\Sigma}_i \boldsymbol{\Sigma}_j)} \left[\frac{1}{n_i} \sum_{k=1}^{n_i} [\mathbf{a}'_{ik} \boldsymbol{\Sigma}_j \mathbf{a}_{ik} - \text{tr}(\boldsymbol{\Sigma}_i \boldsymbol{\Sigma}_j)] + \frac{1}{n_j} \sum_{l=1}^{n_j} [\mathbf{a}'_{jl} \boldsymbol{\Sigma}_i \mathbf{a}_{jl} - \text{tr}(\boldsymbol{\Sigma}_i \boldsymbol{\Sigma}_j)] \right]
\end{aligned}$$

(van der Vaart, 1998, p. 165) with $E(\hat{U}_{n_i n_j}) = 0$ and

$$\text{Var}(\hat{U}_{n_i n_j}) = 2 \left(\frac{1}{n_i} + \frac{1}{n_j} \right) \frac{\text{tr}(\boldsymbol{\Sigma}_i \boldsymbol{\Sigma}_j)^2}{[\text{tr}(\boldsymbol{\Sigma}_i \boldsymbol{\Sigma}_j)]^2},$$

bounded in p . The proof follows by treating each component as for B_i .

B3. Proof of theorem 3

The probability convergence of C_1 , C_2 follows trivially by Chebychev's inequality. Then, the distributions of T_a and T_b under H_1 follow from equation (53) with variance in equation (18) because ξ 's of the normalized kernel $C_3/\text{tr}(\Sigma^2)$ are uniformly bounded away from 0 and ∞ .

Under H_0 , $\xi_{10} = 0 = \xi_{01} = \xi_{20} = \xi_{02}$ in $\text{Var}(C_3/p)$ so that the kernel is degenerate; the non-zero term, $2/n_1 n_2$, that gives the finite variance comes from $\xi_{11} = \text{Var}(h_{11}/p)$. Let λ_m be the eigenvalues of $\tilde{h}_{11}(\cdot) = h_{11}(\cdot)/p - 1$ with $E(\tilde{h}) = 0$, $\text{Var}(\tilde{h}) < \infty$ so that $h(\cdot)$ is square-integrable. Let $\mathbf{X}_{ik} \in \mathbb{R}^p$, $k = 1, \dots, n_i$, $i = 1, \dots, g$, be generated by a probability space $(\mathcal{H}, \mathcal{A}, \mathcal{P})$ with an appropriate Borel σ -algebra and associated probability measure, $\mathcal{P}$, so that $h(\cdot) : \mathbb{R}^m \rightarrow \mathbb{R}$ is a symmetric measurable function and being square-integrable, $h(\cdot) \in \mathcal{L}_2(\mathcal{P})$, where $\mathcal{L}_2(\mathcal{P})$ denotes the space of all square-integrable real random variables with respect to $\mathcal{P}_i$. With kernels composed of bilinear forms, we can assume the space to be Hilbert (Lee, 1990, Ch.3).

Then, under degeneracy, $h(\cdot)$ is a Hilbert–Schmidt kernel that admits an orthonormal decomposition to which the kernel converges in mean square (Lee, 1990, Ch. 3). From Koroljuk and Borovskikh (1994, p. 158), it follows that the limit of T_g under the null, based on $h_{11}(\cdot)$, is that of $\sqrt{2/n_1 n_2} z_{11}$ with $z_{11} \xrightarrow{D} \sum_{m=1}^{\infty} \lambda_m z_{1m} z_{2m}$, as $n_i, p \rightarrow \infty$, where z_{im} are independent sequences of iid $N(0, 1)$ variables. Finally, with the mean of limiting distribution 0 and variance $\sum_{m=1}^{\infty} \lambda_m^2 = 2/n_1 n_2$, the normal limit of $\sqrt{n_1 n_2/2} T_g$ follows as well.

B4. Proof of theorem 4

Here, $\Sigma_i = \Sigma \forall i$. Unbiasedness is trivial. Now, $\text{Var}(C_1)$ follows from $\text{Var}(B_i)$ by independence that gives $\text{Var}(B_i B_j)$, which, in turn, gives $\text{Var}(C_2)$ as

$$\begin{aligned} \text{Var}(C_2) &= \frac{1}{g^2(g-1)^2} \text{Var} \left(\sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g B_i B_j \right) \\ &= \frac{1}{g^2(g-1)^2} \left(2 \sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g \text{Var}(B_i B_j) + 4 \sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g \sum_{\substack{j'=1 \\ j \neq j'}}^g \text{Cov}(B_i B_j, B_i B_{j'}) \right) \\ &= \frac{1}{g^2(g-1)^2} \left(2 \sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g [\text{Var}(B_i) \text{Var}(B_j) + [\text{tr}(\Sigma)]^2 \{\text{Var}(B_i) + \text{Var}(B_j)\}] \right. \\ &\quad \left. + 4(g-1)(g-2) \sum_{i=1}^g [\text{tr}(\Sigma)]^2 \text{Var}(B_i) \right). \end{aligned}$$

Note that in $\text{Var}(C_2)$, $\text{Var}(B_i B_j)$ involves terms like $\text{Cov}[(B_i B_j, B_{i'} B_{j'})]$. There are $g(g-1)(g-2)(g-3)$ covariances that vanish when $i \neq i', j \neq j'$. Further, there are $2g(g-1)(g-2)$ non-zero covariances when $i = i', j \neq j'$ and when $i \neq i', j = j'$, which reduce to the same expression by symmetry. Then,

$$\text{Cov}(B_i B_j, B_i B_{j'}) = E(B_i B_j B_i B_{j'}) - E(B_i B_j) E(B_i B_{j'}) = \text{Var}(B_i) [E(B_j)]^2.$$

Substitution gives the required result. Now, with $P^* = \sum_{i=1}^g \sum_{j=1, j \neq i}^g P(n_i, n_j)$,

$$\text{Var}(C_3) = \frac{1}{P^{*2}} \left(2 \sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g [P(n_i, n_j)]^2 \text{Var}(B_{ij}) + 4 \sum_{i=1}^g \sum_{\substack{j=1 \\ i \neq j}}^g \sum_{\substack{j'=1 \\ i \neq j \neq j'}}^g P(n_i, n_j) P(n_i, n_{j'}) \text{Cov}(B_{ij}, B_{ij'}) \right),$$

with $\text{Cov}(B_{ij}, B_{ij'}) = E(B_{ij} B_{ij'}) - [\text{tr}(\Sigma^2)]^2$. Using B_{ij} and trace properties,

$$E(A_{ijkrls}^2 A_{ij'krls}^2) = E[(\mathbf{D}'_{ikr} \mathbf{D}_{jls})^2 (\mathbf{D}'_{ikr} \mathbf{D}_{j'ls})^2] = 4E(\mathbf{D}'_{ikr} \Sigma \mathbf{D}_{ikr})^2,$$

so that $E(A_{ijkrls}^2 A_{ij'krls}^2) = 16[2\text{tr}(\Sigma^4) + [\text{tr}(\Sigma^2)]^2] = E(B_{ij} B_{ij'})$ because

$$E(B_{ij} B_{ij'}) = \frac{1}{P(n_i, n_j) P(n_i, n_{j'})} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{\substack{r=1 \\ l \neq s}}^{n_i} \sum_{\substack{l=1 \\ l \neq s}}^{n_j} \sum_{\substack{s=1 \\ l \neq s}}^{n_j} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{\substack{r=1 \\ l \neq s}}^{n_i} \sum_{\substack{l=1 \\ l \neq s}}^{n_{j'}} \sum_{\substack{s=1 \\ l \neq s}}^{n_{j'}} E(A_{ijkrls}^2 A_{ij'krls}^2)$$

and the result follows by substitution. For covariances, consider

$$\text{Cov}(C_1, C_2) = \frac{1}{g(g-1) \sum_{i=1}^g Q(n_i)} \sum_{i=1}^g \sum_{j=1, j \neq j'}^g \sum_{j'=1}^g Q(n_i) \text{Cov}(B_i, B_j B_{j'}),$$

which clearly vanishes when $i \neq j \neq j'$. For $i = j \neq j'$, we have

$$\text{Cov}(B_i, B_i B_{j'}) = E(B_i^2) E(B_{j'}) - [E(B_i)]^2 E(B_{j'}) = \text{Var}(B_i) E(B_{j'}),$$

with the same result when $i = j' \neq j$, so that $\text{Cov}(C_1, C_2)$ simplifies to

$$\text{Cov}(C_1, C_2) = \frac{4n}{g \sum_{i=1}^g Q(n_i)} \text{tr}(\Sigma^2) \text{tr}(\Sigma),$$

with $n = \sum_{i=1}^g n_i$, as required. Now, with $S^* = \sum_{i=1}^g Q(n_i) \sum_{j=1, j \neq j'}^g \sum_{j'=1}^g P(n_j, n_{j'})$,

$$\text{Cov}(C_1, C_3) = \frac{1}{S^*} \sum_{i=1}^g \sum_{\substack{j=1 \\ j \neq j'}}^g \sum_{j'=1}^g Q(n_i) P(n_j, n_{j'}) \text{Cov}(B_i, B_{jj'}),$$

where

$$\text{Cov}(B_i, B_{jj'}) = \frac{1}{8Q(n_i) P(n_j, n_{j'})} \sum_{\substack{k=1 \\ k \neq r}}^{n_i} \sum_{\substack{r=1 \\ l \neq s}}^{n_i} \sum_{\substack{l=1 \\ l \neq s}}^{n_j} \sum_{\substack{s=1 \\ l' \neq s'}}^{n_j} \sum_{\substack{l'=1 \\ l' \neq s'}}^{n_{j'}} \sum_{\substack{s'=1 \\ l' \neq s'}}^{n_{j'}} \text{Cov}(A_{ikr}, A_{jj'ls'l's'}^2),$$

which vanishes when $i \neq j \neq j'$. For $i = j \neq j'$, we have

$$E(A_{ikr}, A_{jj'ls'l's'})^2 = E \left[\mathbf{D}'_{ikr} \mathbf{D}_{ikr} (\mathbf{D}'_{jls} \mathbf{D}_{j'l's'})^2 \right] = E (\mathbf{D}'_{ikr} \mathbf{D}_{ikr} \mathbf{D}'_{ikr} 2\mathbf{\Sigma} \mathbf{D}_{ikr}).$$

By equation (64), $\text{Cov}(A_{ikr}, A_{jj'ls'l's'}) = 16\text{tr}(\mathbf{\Sigma}^3)$ and same for $i = j' \neq j'$. This simplifies $\text{Cov}(B_i, B_{jj'})$ to $2\text{tr}(\mathbf{\Sigma}^3)/Q(n_i)$, which gives $\text{Cov}(C_1, C_3)$ as required. Finally, with $P^* = \sum_{j=1}^g \sum_{j'=1}^g P(n_{i'}, n_{j'})$,

$$\text{Cov}(C_2, C_3) = \frac{1}{g(g-1)P^*} \sum_{i=1}^g \sum_{j=1}^g \sum_{i' \neq j}^g \sum_{j'=1}^g \text{Cov}(B_i B_j, B_{i'j'}),$$

with

$$\begin{aligned} \text{Cov}(B_i B_j, B_{i'j'}) &= \frac{1}{16Q(n_i)Q(n_j)P(n_{i'}, n_{j'})} \\ &\sum_{k=1}^{n_i} \sum_{r=1}^{n_i} \sum_{l=1}^{n_j} \sum_{s=1}^{n_j} \sum_{k' \neq r}^{n_{i'}} \sum_{l' \neq s}^{n_{j'}} \sum_{r'=1}^{n_{i'}} \sum_{s'=1}^{n_{j'}} \text{Cov}(A_{ikr} A_{jls}, A_{i'j'k'r'l's'}^2), \end{aligned}$$

which vanishes when $i \neq i', j \neq j'$. For $i = i', j = j'$, we have

$$\text{Cov}(A_{ikr} A_{jls}, A_{i'j'k'r'l's'}^2) = 64\text{tr}(\mathbf{\Sigma}^4) + 64\text{tr}(\mathbf{\Sigma}^3)\text{tr}(\mathbf{\Sigma}),$$

following equation (26) in Ahmad & Rosen (2015). This gives

$$\begin{aligned} \text{Cov}(B_i B_j, B_{ij}) &= \frac{1}{16Q(n_i)Q(n_j)P(n_i, n_j)} \sum_{k=1}^{n_i} \sum_{r=1}^{n_i} \sum_{l=1}^{n_j} \sum_{s=1}^{n_j} 64[\text{tr}(\mathbf{\Sigma}^4) + \text{tr}(\mathbf{\Sigma}^3)\text{tr}(\mathbf{\Sigma})] \\ &= \frac{4}{P(n_i, n_j)} [\text{tr}(\mathbf{\Sigma}^4) + \text{tr}(\mathbf{\Sigma}^3)\text{tr}(\mathbf{\Sigma})]. \end{aligned}$$

For $i = i', j \neq j'$, we have, using again trace properties,

$$\begin{aligned} E(A_{ikr} A_{jls} A_{i'j'k'r'l's'}^2) &= E [\mathbf{D}'_{ikr} 2\mathbf{\Sigma} \mathbf{D}_{ikr} \mathbf{D}'_{i'j'k'r'l's'} \mathbf{D}_{i'j'k'r'l's'}] \\ &= 32\text{tr}(\mathbf{\Sigma}^3)\text{tr}(\mathbf{\Sigma}) + 16\text{tr}(\mathbf{\Sigma}^2)[\text{tr}(\mathbf{\Sigma})]^2, \end{aligned}$$

with $\text{Cov}(A_{ikr} A_{jls}, A_{i'j'k'r'l's'}^2) = 32\text{tr}(\mathbf{\Sigma}^3)\text{tr}(\mathbf{\Sigma})$, and same if $i \neq i', j = j'$. Now,

$$\begin{aligned} \text{Cov}(B_i B_j, B_{ij'}) &= \frac{1}{16Q(n_i)Q(n_j)P(n_i, n_{j'})} \sum_{k=1}^{n_i} \sum_{r=1}^{n_i} \sum_{l=1}^{n_j} \sum_{s=1}^{n_j} \sum_{k' \neq r}^{n_{i'}} \sum_{l' \neq s}^{n_{j'}} \sum_{r'=1}^{n_{i'}} \sum_{s'=1}^{n_{j'}} 32\text{tr}(\mathbf{\Sigma}^3)\text{tr}(\mathbf{\Sigma}) \\ &= \frac{2}{Q(n_i)} \text{tr}(\mathbf{\Sigma}^3)\text{tr}(\mathbf{\Sigma}). \end{aligned}$$

Similarly, for $i \neq i', j = j'$, $\text{Cov}(B_i B_j, B_{i'j}) = \frac{2}{Q(n_j)} \text{tr}(\Sigma^3) \text{tr}(\Sigma)$. Substitution gives, with $P^* = \sum_{j=1}^g \sum_{\substack{j'=1 \\ j \neq j'}}^g P(n_i n_{j'})$,

$$\begin{aligned} \text{Cov}(C_2, C_3) &= \frac{1}{g(g-1)P^*} \left(\sum_{\substack{i=1 \\ i \neq j}}^g \sum_{j=1}^g P(n_i, n_j) \frac{4}{P(n_i, n_j)} [\text{tr}(\Sigma^4) + \text{tr}(\Sigma^3) \text{tr}(\Sigma)] \right. \\ &\quad + 2 \sum_{i=1}^g \sum_{j=1}^g \sum_{\substack{j'=1 \\ i \neq j \neq j'}}^g P(n_i n_{j'}) \frac{2}{Q(n_i)} \text{tr}(\Sigma^3) \text{tr}(\Sigma) \\ &\quad \left. + 2 \sum_{i=1}^g \sum_{j=1}^g \sum_{\substack{i'=1 \\ i \neq j \neq i'}}^g P(n_i n_{j'}) \frac{2}{Q(n_j)} \text{tr}(\Sigma^3) \text{tr}(\Sigma) \right) \\ &= \frac{1}{g(g-1)P^*} \left(8g(g-1) [\text{tr}(\Sigma^4) + \text{tr}(\Sigma^3) \text{tr}(\Sigma)] \right. \\ &\quad \left. + 4g(g-1) \text{tr}(\Sigma^3) \text{tr}(\Sigma) \left[\sum_{j'=1}^g Q(n_{j'}) + \sum_{i'=1}^g Q(n_{i'}) \right] \right) \\ &= \frac{8}{P^*} \left(\text{tr}(\Sigma^4) + \text{tr}(\Sigma^3) \text{tr}(\Sigma) + \text{tr}(\Sigma^3) \text{tr}(\Sigma) \sum_{i=1}^g Q(n_i) \right). \end{aligned}$$

B5. Proof of theorem 7

This proof follows on the same lines as that under normality, except replacing the moments of quadratic and bilinear forms under non-normal model. As the new moments differ from those under normality by adding components involving M 's, which eventually vanish under assumption 5, the final asymptotic results are the same as under normality. Consider B_i . From the proof of theorem 2, $\hat{U}_{n1} = \frac{m}{n_1} \sum_{k=1}^{n_1} g(\mathbf{x}_{1k})$, with $g(\mathbf{x}_{1k}) = \tilde{h}_1(\mathbf{x}_{1k}) = [\mathbf{a}'_{1k} \Sigma_1 \mathbf{a}_{1k} + \text{tr}(\Sigma_1^2)] / 2\text{tr}(\Sigma_1^2) - 1$, where, using now the results from the proof of theorem 7, we obtain

$$\text{Var}[g(\mathbf{x}_{1k})] = \frac{2\text{tr}(\Sigma_1^4) + M_2}{4[\text{tr}(\Sigma_1^2)]^2},$$

and, under assumption 5, the second component involving M_2 vanishes leaving the asymptotic variance the same as that under normality. The rest of the proof then follows exactly as under normality and similarly also the proof of B_{ij} .